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EDUCATION

2009 - 2013

Acharya institute of technology,
Bangalore

- Bachelor of Engineering in
Mechanical

CORE COMPETENCIES

- Data Science & Analysis
- Machine Learning / Deep Learning
- Predictive Modeling
- Data Mining
- Statistical Analysis
- Natural Language Processing
- Generative AI / LLM

TECHNICAL SKILLS

- Python Programming
- Natural Language Processing
- Vector database
- Power BI & Tableau
- Azure DevOps / Git
- Linux/Redhat/SQL

LANGUAGES

- English
- Kannada
- Hindi (Basics)

KIRAN G

AI/ML Specialist | Machine Learning | NLP | Generative AI

PROFILE SUMMARY

- Data scientist with over 10 years of total experience and with 6 years of experience in delivering valuable insights through data analytics and machine learning driven modelling solutions to challenging business problems across Automotive, Manufacturing, Banking, Insurance, and CPG domains.
- Knowledgeable with machine learning (Supervised Learning and Unsupervised Learning algorithms), NLP using Generative AI with LLM, and Knowledge Graph.
- Data Science enthusiasts with excellent communication skills - flexible/quick learners comfortable with variety of platforms – familiar with new age development frameworks.
- Devise POCs on Search Engine, Question Answering system (on enterprise data) using both OpenAI and open-sourced based LLM models and deploying them to production.
- I am good at developing and maintaining code, implementing efficient CI-CD pipelines, testing new features, and resolving issues promptly. I have consistently delivered high-quality code with a keen focus on minimizing defects.

WORK EXPERIENCE

- Accenture** 2022 - PRESENT
AI Engineering Senior Analyst
 - Devise POCs on Search Engine, Question Answering system (on enterprise data) using both OpenAI and open-sourced based LLM models and deploying them to production.
 - Construct end-to-end pipelines starting from data acquisition to inference in the cloud.
 - Experienced in CI/CD pipelines, release management process and security compliances, and DevOps principles
- TCS** 2020 - 2022
Associate Data Scientist
 - Design and develop AI solutions using machine learning and deep learning algorithms.
 - Build end-to-end pipelines from data pre-processing till model inference and monitoring in GCP.
 - Monitor performance of existing models, in production, for data drift and concept drift.
- CSG** 2014 - 2020
Analyst
 - Devised POCs for various process of vehicle crash analysis like Predict rollover test speed, Loads predictions for "General Motors"
 - Deploy automation script to reduce the repeated tasks by automating the process of CAE modelling and connection activities.

1) SCALABLE HYBRID SEARCH ENGINE FOR ENTERPRISE DATA

- **Objective:** Replace legacy keyword search with a high-recall, low-latency hybrid search engine enhanced with generative AI capabilities.
- **Solution & Responsibilities:**
 - Built an ETL pipeline to extract content from unstructured files (PDF, PPTX, DOCX, XLSX), preprocess text, generate embeddings using OpenAI text-embedding-ada-002, and index in Elastic Search (vector + BM25).
 - Developed a unified search API using FastAPI, deployed via Azure App Service, and scheduled data refresh pipelines with Azure Functions.
 - Implemented reciprocal rank fusion (RRF) to merge results from vector and keyword retrieval.
 - Integrated GPT-4o (Azure OpenAI) for on-the-fly summarization of top retrieved documents to improve result comprehension and user engagement.
- **Tech Stack:** Python, FastAPI, Azure App Service, Azure Functions, Azure DevOps, Docker, Elastic Search (BM25 + Vector), Azure OpenAI (text-embedding-ada-002, GPT-4o).
- **LLM Technique:** Hybrid retrieval with RRF + GPT-4o summarization.
- **Impact:**
 - Used by 3 000+ practitioners; avg. 120 min/week saved per user.
 - GPT-based summaries improved result interpretability and decision-making.
 - Outperformed traditional search in Top-5 and Top-10 Precision metrics; user feedback indicated significant improvement in relevance and usability.

2) RESPONSIBLE RAG SYSTEM WITH QA, SUMMARIZATION & LANGUAGE GUARDRAILS

- **Objective:** Develop a secure and trustworthy enterprise AI assistant that combines question answering, summarization, and Responsible AI enforcement using Generative AI.
- **Solution & Responsibilities:**
 - Designed a Retrieval-Augmented Generation (RAG) pipeline to retrieve relevant enterprise content from Milvus Vector DB, and generate contextual answers using GPT-4o (Azure OpenAI).
 - Enabled automatic summarization of top-ranked documents using GPT-4o when users requested a summary or entered open-ended queries.
 - Integrated guardrails for Responsible AI using the guardrails-ai library, including:
 - Detection and mitigation of toxicity, bias, and unsafe content
 - Enforced English-only output constraint across all LLM generations
 - Schema validation for summaries and answers to ensure consistency and structure
 - Developed a modular backend with FastAPI, and used PostgreSQL to log user queries, feedback, model responses, and guardrail violation events.
 - Implemented continuous monitoring and audit logging for model behavior, used for iterative retriever threshold tuning and safe prompt refinement.
- **Tech Stack:** Python, FastAPI, Milvus, PostgreSQL, Azure OpenAI (GPT-4o), guardrails-ai, Docker, Azure DevOps
- **LLM Techniques:**
 - RAG (Milvus vector search + GPT-4o completion)
 - Generative Summarization (on top-k retrieved chunks)
 - Guardrails for responsible AI enforcement (toxicity filtering, language policy, structured outputs)
 - Multi-turn QA with context-aware prompt chaining
- **Impact:**
 - Ensured 100% English-only output compliance across diverse input scenarios
 - Filtered or blocked 98.7% of unsafe outputs before user delivery
 - Boosted user trust and adoption, with a 4.5/5 satisfaction rating

3) AGENTIC INSIGHT GENERATOR FOR CLIENT EARNINGS TRANSCRIPTS

- **Objective:** Build an intelligent agentic system that automatically analyzes earnings transcripts (e.g., ManpowerGroup Inc.), extracts high-impact strategic themes, classifies them into actionable consulting areas, and delivers tailored solution insights to clients via email and presentation decks.
- **Solution & Responsibilities:**
 - Designed a multi-agent workflow using a centralized RAG pipeline to:
 - Ingest and chunk earnings call transcripts
 - Classify themes into categories of Growth, Cost, or Sustainability using a structured prompt and reasoning chain using GPT-4o (Azure OpenAI)

- For each theme, the system generates:
 - Theme Name
 - ThemeCategorization based on business impact
 - ThemeBrief (≤ 300 characters)
 - ValidatingFactsfromEarningsTranscript — pulled directly from transcript with citations
- Each theme is passed as a semantic query to an internal enterprise solution knowledge base indexed in Milvus Vector DB, retrieving:
 - Relevant solution components, prebuilt narratives, and case studies which will help improve the client performance.
- Implemented Guardrails-AI to:
 - Enforce structured schema for theme response (JSON format)
 - Ensure outputs are in English only
 - Detect hallucinated or unsupported claims from transcripts
- Developed automated flows to:
 - Send customized emails to client stakeholders with insights + solutions
 - Offer an option to auto-generate a client-facing PowerPoint deck for internal/external sales use.
- **Tech Stack:** Python, FastAPI, GPT-4o (Azure OpenAI), Milvus, Haystack framework, CrewAI, guardrails-ai, PostgreSQL, PowerPoint generation (python-pptx)
- **LLM Technique:**
 - RAG for solution retrieval
 - Agentic orchestration of multiple LLM tasks (extraction of theme and theme brief → classification of theme category → retrieval from enterprise solution → sending email to client → powerpoint generation)
 - Structured prompt chaining + validation with Guardrails
- **Impact:**
 - Reduced analyst time for earnings transcript review by 90%
 - Enabled scalable and consistent client insight generation across 50+ enterprise clients
 - Increased solution pitch conversion by 25% due to aligned and personalized recommendations.

Cloud Certifications

Microsoft Certification:

1. Azure AI Fundamentals - AI 900
2. Azure AI Engineer Associate - AI102

AWS certification:

1. AWS Certified Machine Learning - Specialty

GCP:

1. GCP Associate Cloud Engineer
2. GCP - Professional Data Engineering
3. GCP - Professional Machine learning Engineer